# Coverage Is Not Predictive Fidelity: A Split-Landscape Analysis of Data Condensation for Gradient-Boosted Trees

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***Abstract*—**
**Data condensation replaces a large training set with a small surrogate, but tabular condensers are usually judged only after retraining a model on each candidate. For histogram gradient-boosted decision trees (GBDTs), we ask whether a candidate preserves the split decisions the full-data booster would make. We introduce Split-Landscape Discrepancy (SLD), a no-candidate-retraining diagnostic that compares full and condensed split-gain landscapes under fixed preprocessing and prediction state.**

**Across 17 condensers on 26 binary tabular datasets, method-level split-regret is strongly rank-correlated with downstream AUROC in the small-to-mid-size regime (Spearman $\rho = -0.87$), making SLD useful for screening bad surrogates and ranking method families, not for single-dataset top-1 selection. The diagnostic also exposes why coverage can fail: a greedy high-gain threshold-coverage selector is among the weakest performers because unweighted coverage differs from gain-weighted structural fidelity. At very large scale, random sampling becomes hard to beat and root-regret loses resolution as reasonable surrogates clear the dominant split margins. We release the benchmark, scripts, processed tables, and diagnostic to support reproducible evaluation.**

***Index Terms*—data condensation; gradient-boosted decision trees; coreset selection; tabular data; no-candidate-retraining diagnostics**

## I. Introduction

ML engineers and data scientists who train gradient-boosted models on tabular data face a recurrent resource tension. A single retraining run on a dataset of hundreds of thousands of rows can consume substantial compute; in our scale audit, condensed retraining reaches 160×–5,734× speedups at the largest training sizes. Repeated runs for hyperparameter search, model versioning, or data-sharing workflows multiply that cost many times over. Condensation is not itself a privacy or compliance guarantee, but a small surrogate can be useful when combined with the appropriate governance or privacy mechanisms. The natural remedy, replacing the full dataset with a small, high-fidelity surrogate, is already practiced informally through random subsampling and stratified sampling. What is missing is a principled way to answer the question that matters most *before* committing to retraining on a surrogate: does this condensed set faithfully preserve the structure that the booster would have learned from the original data?

The dataset condensation and coreset-selection literature has produced a rich supply of methods to construct small surrogates [1]–[4]. For tabular data specifically, several methods have recently been adapted or purpose-built for non-image settings [5], [6]. Yet the evaluation paradigm is almost universally the same: train a fresh model on the condensed set, measure downstream predictive performance, and compare. This protocol has two practical shortcomings. First, it requires precisely the retraining one hoped to avoid. Second, it collapses a mechanistic question, namely *why* a condensed set works or fails, into a scalar outcome that gives no guidance for debugging or selection.

Gradient-boosted decision trees [7]–[9] offer a distinctive opportunity to break this cycle. A GBDT tree structure is determined by split-selection decisions: at each node, the algorithm chooses the feature and threshold that maximize the regularized gain criterion (Section III). Predictive behaviour also depends on leaf values, learning rate, boosting rounds, missing-value handling, and regularization. The gain criterion is histogram-sufficient once first- and second-order loss statistics are fixed; those statistics may come from a root base score or from a reference booster. Comparing the histogram landscapes of the full and condensed datasets is therefore a no-candidate-retraining screening diagnostic, not a claim that no model state is ever used.

We introduce the *Split-Landscape Discrepancy* (SLD), which formalizes this comparison as an $\ell_p$ distance between the split-gain landscapes of a full dataset and its surrogate (Section III). Using SLD as a measurement instrument, we conduct a systematic comparative evaluation of 17 condensation methods, spanning classical coresets, gradient-based selectors, distribution-matching distillation, and gain-path distillation, under a unified protocol across a broad collection of tabular binary classification datasets. The analysis is a measurement study; we do not propose a new condensation method and we do not claim a new state of the art.

Our analysis surfaces three empirical findings of practical consequence. At the method-mean level, split-regret strongly ranks method families by downstream AUROC in the small-to-mid-size benchmark, enabling screening before retraining on every candidate. A provably coverage-optimal selector is simultaneously among the weakest methods by predictive performance, a demonstration that coverage and structural fidelity are independent objectives. At deployment scale, random sampling becomes difficult to beat, and SLD is better read as a rejection signal than as a single-best selector.

**Contributions.**

- **The SLD diagnostic.** We introduce the Split-Landscape Discrepancy, a no-candidate-retraining metric that quantifies how faithfully a condensed dataset reproduces the GBDT split-gain landscape of the full data under a fixed prediction state (Section III).
- **Systematic comparative analysis.** We present a 17-method evaluation specifically for GBDT-on-tabular settings, pooling results over thousands of dataset $\times$ budget $\times$ seed configurations and measuring both downstream predictive performance and landscape fidelity (Sections IV and V).
- **Three empirically grounded insights.** A single landscape score is method-level rank-correlated with AUROC at Spearman $\rho = -0.87$ in the small-to-mid-size benchmark; within-dataset trends are generally negative after controlling for dataset scale. Coverage and predictive fidelity are decoupled: a $(1-1/e)$-optimal coverage selector is among the weakest downstream performers, explained by a reference-chain decomposition into structure and leaf-estimate gaps. Structure is robust to perturbation below a margin-linked threshold; the dose is a fraction of the split margin, not a fraction of intentionally corrupted splits (Section V).
- **Scope boundaries.** We document where the lens does not extend: regression tasks (structure-dominated failure), hyperparameter-rank transfer (near-zero Jaccard), and no-retraining selection of the single best method (top-3 rate 0.44), measured under the same protocol (Section VI).
- **Released artifacts.** The benchmark protocol, SLD implementation, processed tables, seeds, command lines, and plotting scripts are released at https://github.com/VIVAN-RANGRA/sld-gbdt-condensation.

## II. Related Work

*a) Dataset distillation and condensation.:* Dataset distillation [1] and its successors optimise a small synthetic set so that a model retrained on it matches one trained on the full data, using gradient matching [2], distribution matching [10], or trajectory matching [11]. Comprehensive surveys [12]–[14] emphasize downstream evaluation, mostly on image benchmarks, and do not provide a GBDT-specific split-landscape fidelity metric. Extending distillation to tabular data has attracted recent attention [5], [15]–[17], but these works retain the retrain-and-evaluate paradigm. Our contribution is orthogonal: rather than a new condensation method, we introduce a no-candidate-retraining diagnostic that measures whether *any* condensed set, regardless of the method that produced it, preserves the GBDT split-gain landscape.

*b) Coreset selection.:* Classical coreset methods select real training examples by geometric or gradient-based criteria: herding [3] matches distribution moments; k-center [18] minimises the maximum coverage distance; CRAIG [4] and GRAD-MATCH [19] match gradient norms via submodular maximisation; EL2N and GraNd [20] use early-training gradient norms. All of these methods were designed for and evaluated on differentiable models. Translating their selection signals to the GBDT setting is non-trivial because GBDTs are not differentiable with respect to the training set; we include adapted variants in our 17-method benchmark and show that their landscape fidelity, measured by SLD, predicts their tabular AUROC. The GBDT-native subsampling methods GOSS [9] and MVS [21] operate per boosting round rather than producing a static condensed set; MVS is the theoretically closest prior work, as it explicitly minimises the variance of the split-scoring estimator. Data valuation [22], [23] and influence-function methods offer related but complementary perspectives on data contribution; they often require retraining on many subsets or differentiable-model assumptions and are not directly applicable in their standard form to histogram GBDTs.

*c) Why GBDTs for tabular data.:* Gradient-boosted decision trees [7]–[9], [24] remain one of the strongest practical model classes for tabular data [25], [26]. This motivates a condensation evaluation framework specific to tree-structured inductive bias. The split-gain criterion [27], [28] aggregates per-bin gradient statistics, and the resulting landscape fully determines which feature and threshold a node selects, a structure that has no direct analogue in differentiable models. Our SLD exploits this structure to build a measurement instrument that is GBDT-mechanism-specific: it explains *why* a condensed set preserves tree behaviour, not merely whether it does.

*d) The $C^2TC$ contrast.:* The most closely related recent work is $C^2$TC [6] (ICDE 2026), a training-free tabular condensation *method*: it optimises class allocation and feature representation via class-adaptive clustering to produce a condensed set efficiently. Our work is fundamentally different in intent. $C^2$TC is a *condenser*: it produces a surrogate dataset. SLD is a *measurement instrument*: it audits the fidelity of any surrogate, including one produced by $C^2$TC, with respect to the GBDT split-gain landscape. Where $C^2$TC is model-agnostic in its clustering objective, SLD is GBDT-mechanism-specific: it reads the landscape from histogram statistics under a fixed prediction state without retraining on each candidate. We do not include $C^2$TC in the full grid because its official implementation was not integrated before our 17-method grid was frozen. As a compatibility audit, we ran the official $C^2TC$ code on our Adult split at budget 200; the generated set obtained AUROC 0.769 and raw $\mathrm{SLD}_\infty = 1228.2$, confirming that SLD can audit $C^2TC$-generated surrogates; this single run does not constitute a full $C^2TC$ baseline.

## III. Background and the SLD Diagnostic

### A. GBDT Split-Gain Landscape

Let the full training dataset be $D = \{(x_i, y_i)\}_{i=1}^N$ with features $x_i \in \mathbb{R}^F$ and labels $y_i$. A gradient-boosted decision tree (GBDT) fits an additive ensemble of trees by performing, at each node, a greedy split that maximises the regularised gain [7], [8]. Given first- and second-order loss statistics $g_i = \partial_{\hat{y}}\ell(\hat{y}_i, y_i)$ and $h_i = \partial^2_{\hat{y}}\ell(\hat{y}_i, y_i)$ for each instance $i$, the gain of partitioning the instance set $I$ at a node into left and right children $I_L, I_R$ is

$$\mathcal{G}(I_L, I_R) = \frac{1}{2}\left[\frac{\left(\sum_{i\in I_L} g_i\right)^2}{\sum_{i\in I_L} h_i + \lambda} + \frac{\left(\sum_{i\in I_R} g_i\right)^2}{\sum_{i\in I_R} h_i + \lambda} - \frac{\left(\sum_{i\in I} g_i\right)^2}{\sum_{i\in I} h_i + \lambda}\right] - \gamma, \tag{1}$$

where $\lambda \geq 0$ is the $\ell_2$ leaf-weight regulariser and $\gamma \geq 0$ is the minimum gain threshold [8]. Modern histogram-based GBDTs [9], [24] discretise each feature $f$ into $B$ bins; per-bin sums of $(g, h)$ suffice to evaluate every candidate split.

**Lemma 1** (Histogram sufficiency; standard, e.g. [9], [21]). *Given the current gradients and Hessians, a histogram GBDT with $B$ bins per feature has its full per-node split-gain landscape determined entirely by the $F \times B$ grid of per-bin $(g, h)$ aggregates. No other information from the training set is required to evaluate* (1) *at every candidate (feature, threshold) pair.*

We define the *split-gain landscape* $\Gamma_D \in \mathbb{R}^{F\times(B-1)}$ as the gain $\mathcal{G}$ that each (feature, bin-boundary threshold) candidate would achieve on dataset $D$ under a fixed prediction state. Lemma 1 is a standard consequence of the histogram GBDT algorithm; we state it here to ground the diagnostic. Its importance is histogram sufficiency: once the prediction state fixes $g_i, h_i$, $\Gamma_D$ can be computed in one histogram pass.

### B. Split-Landscape Discrepancy

Let $D' = \{(x'_j, y'_j, w'_j)\}_{j=1}^m$ be a weighted selected or synthetic surrogate of $D$, with $m \ll N$. Synthetic points are binned using the same feature preprocessing and bin boundaries as $D$; their labels may be hard labels or soft labels converted to the objective's expected gradients. We quantify the structural gap between $D$ and $D'$ via their landscape distance.

**Definition 1** (Split-Landscape Discrepancy). *Let the raw landscape error be $\delta_p(D, D') = \|\Gamma_D - \Gamma_{D'}\|_p$. The relative Split-Landscape Discrepancy of order $p$ between dataset $D$ and surrogate $D'$ is*

$$\mathrm{SLD}_p(D, D') \;=\; \frac{\delta_p(D, D')}{\max(\|\Gamma_D\|_p, \epsilon)}, \tag{2}$$

*using the same bins, objective, regularisation, base score, and feature preprocessing for $D$ and $D'$. We use $p = \infty$ (worst-case relative split distortion) and the rank-based* split-regret *variant. Root-SLD compares the root landscape. Reference-tree early-node SLD averages over early nodes of a fixed full-data reference tree, using the reference node memberships. Full-path SLD extends the same comparison along deeper reference-tree paths.*

Both variants are computed from histogram statistics under a fixed prediction state, and crucially *no model is retrained on $D'$*. Figure 1 illustrates how SLD fits into the condensation evaluation pipeline as a no-candidate-retraining alternative to the retrain-and-evaluate loop.

### C. Diagnostic Sanity Checks

The diagnostic rests on two standard facts about histogram GBDTs and set coverage. We use them only to motivate what SLD measures; the contribution of the paper is the empirical measurement study across 17 condensation methods.

*a) Margin preservation.:* Let $\Delta_v$ be the gain margin between the best-ranked and second-best-ranked split at node $v$ on $D$. If

$$\delta_\infty(D, D') < \Delta_v/2, \tag{3}$$

then the argmax split selected by $D'$ coincides with the argmax selected by $D$ at node $v$, assuming a unique argmax and deterministic tie-breaking. In the dose-response cells satisfying this raw-error margin condition, root-level split agreement equals 1.0 (Section V-C).

*b) Coverage guarantee.:* The split-coverage objective is the unweighted fraction of full-data high-gain split thresholds whose adjacent bins are represented by at least one selected or weighted surrogate point. It is binary threshold coverage, not gain-weighted path fidelity. As a standard set-coverage objective, it is monotone submodular under a cardinality constraint, so greedy maximisation has the usual $(1 - 1/e)$ approximation guarantee [29]. This defines the HistDistill-Greedy selector: it achieves near-optimal coverage of the split landscape, yet performs poorly by downstream predictive performance (Section V-B). The gap between coverage and predictive fidelity is the central empirical finding of Section V-B.

### D. Computing SLD: Algorithm

Algorithm 1 summarises the SLD computation. The key property is that, after the fixed prediction state is available, scoring a candidate requires only one histogram pass and one arithmetic comparison of landscapes; no candidate GBDT is trained. The pass costs $O(NFB)$ to build full-data histograms and $O(mFB)$ for the surrogate, with shared bins and $O(FB)$ landscape memory per node.

**Algorithm 1** Compute Split-Landscape Discrepancy without candidate retraining

**Require:** Full dataset $D$, weighted surrogate $D'$, bins, norm order $p$, fixed prediction state $\pi$

**Ensure:** $\mathrm{SLD}_p(D, D')$ and split-regret score

1: Compute first- and second-order loss statistics for $D$ under $\pi$ $\triangleright$ root base score or reference-booster predictions

2: Build histogram $H_D \in \mathbb{R}^{F\times B\times 2}$: for each feature $f$ and bin $b$, store $\left(\sum_{i:x_{if}\in b} g_i,\ \sum_{i:x_{if}\in b} h_i\right)$

3: Compute landscape $\Gamma_D \in \mathbb{R}^{F\times(B-1)}$ via (1) applied to every candidate split

4: Repeat Steps 1–3 for $D'$ to obtain $\Gamma_{D'}$ $\triangleright$ selected rows reuse weights; synthetic rows are evaluated under $\pi$

5: $\mathrm{SLD}_p \leftarrow \|\Gamma_D - \Gamma_{D'}\|_p / \max(\|\Gamma_D\|_p, \epsilon)$

6: **Split-regret:** let $(f^*, b^*) = \arg\max_{f,b} [\Gamma_D]_{f,b}$ and $(\hat{f}, \hat{b}) = \arg\max_{f,b} [\Gamma_{D'}]_{f,b}$;

7: split-regret $\leftarrow \max\left(0, \frac{[\Gamma_D]_{f^*,b^*} - [\Gamma_D]_{\hat{f},\hat{b}}}{\max(|[\Gamma_D]_{f^*,b^*}|, \epsilon)}\right)$

8: **return** $\mathrm{SLD}_p$, split-regret

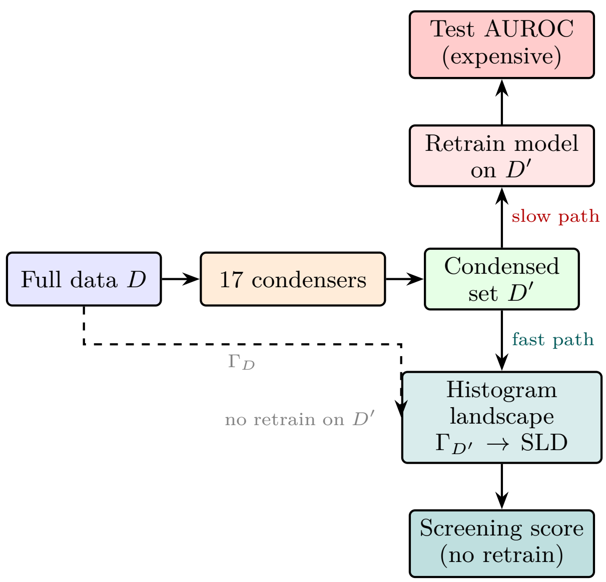


Fig. 1: **SLD evaluation pipeline.** After selecting a condensed set $D'$ from full data $D$ via any of 17 condensers, the fast path computes the candidate's histogram split-gain landscape $\Gamma_{D'}$ under the same fixed prediction state as $\Gamma_D$. It provides a cheap screening signal before retraining on each candidate, in contrast to the slow path of retrain-and-test evaluation.

## IV. Experimental Setup

### A. Methods Evaluated

We evaluate 17 condensation methods spanning five groups; Table I gives the full taxonomy. **Geometry/classical coresets** include Random sampling, Herding [3], and K-center [18]. **GBDT-native subsampling** methods include GOSS [9] and MVS [21], adapted to produce static condensed sets. **Gradient- and score-based selection** methods include GraNd, EL2N [20], CRAIG-style [4], GradMatch-style [19], and gradient sampling (loss-weighted). **Distribution-matching distillation** is represented by DistributionMatching [10] adapted to tabular features. **Gain/landscape-based methods** form our primary family: HistDistill-Greedy (the $(1-1/e)$-optimal coverage selector), HistDistill-Refined and HistDistill-Density (the practical variants optimising landscape fidelity directly), gain sketch, gain-path sketch, and the legacy gain-path refined method. The last three are prior gain-path methods against which we benchmark the refined family.

### B. Datasets and Protocol

Our experiments use tabular datasets drawn from OpenML [30], PMLB [31], and UCI-sourced public datasets; the full inventory appears in Table II. The artifact repository lists the exact OpenML/PMLB identifiers and processed split checksums. The core study is on binary classification. For each dataset we condense the training set to five budgets, $\{10, 25, 50, 100, 200\}$ rows, and repeat every (dataset, method, budget) configuration over five random seeds. Across the 26 binary datasets and the five core methods this gives 3,250 retrained-model evaluations for headline performance (Table III) and 650 landscape-fidelity measurements, one per dataset, method, and budget.

The broader 17-method landscape audit uses a lighter but complete diagnostic grid: 26 datasets, two audited budgets (25 and 50 rows), and three seeds, for 156 dataset–budget–seed cells per method. We use this grid for Figure 2 and the method-level ordering analyses. The decomposition study in Table VI is a diagnostic subset used only to illustrate the two reference-chain failure modes, so its absolute AUROCs should not be compared directly to the pooled headline values.

Three further studies probe the diagnostic beyond this grid. The robustness study injects calibrated split-gain noise at eight dose levels on all 26 binary datasets, with three seeds and three repeats per level, for 1,872 measurements. Generality studies add eight multiclass datasets (one-vs-rest macro-AUROC), eight regression datasets ($R^2$), and a twelve-dataset hyperparameter-transfer benchmark; Section VI gives the protocol details and outcomes. A separate scale audit uses Covertype, SUSY, and HIGGS at training sizes from 20,000 to 10.5M rows with 400-row condensed sets. This audit tests runtime scaling and whether the split-regret ordering survives at deployment scale; it is not used to tune the main 26-dataset benchmark.

### C. Metrics

**Downstream predictive performance.** We score binary tasks by AUROC, multiclass tasks by one-vs-rest macro-AUROC, and regression tasks by $R^2$. Every score is computed on a held-out test set with the model retrained on the condensed set.

**Fidelity.** (2) defines the relative $\ell_\infty$ landscape distance, while split-regret is the normalised gain lost when a node follows the surrogate's argmax split rather than the full-data one. The reported SLD columns and Figure 3 use raw $\ell_\infty$ landscape error to expose the scale confounding in Finding 2; these raw values are not cross-dataset scores. Split-regret is the primary normalised ranking metric. Both are read off the histogram statistics before retraining on any candidate surrogate.

**Error decomposition.** To separate the two ways a condensed set can hurt a tree, we evaluate three models in a reference chain. The *structure reference* uses the full-data tree structure with leaves refit under the decomposition protocol. The *leaf-refit model* uses the condensed-set tree structure but refits its leaves on full data, isolating the cost of the structure alone. The *deployed model* is trained only from the condensed set. We define structure error as reference AUROC minus leaf-refit AUROC, and leaf-estimate gap as leaf-refit AUROC minus deployed AUROC. These two gaps add along the reference-to-deployed chain; they are not intended to sum to the ordinary full-data AUROC gap. We also track how representatively the condensed set fills the model's leaves, defined as the $\ell_1$ distance between the condensed- and full-data distributions of training instances across the leaves; we call this the leaf-population mismatch.

### D. Learner and Hardware

The reference landscape is built with XGBoost hist trees: 80 boosting rounds, max depth 3, learning rate 0.08, $\lambda = 1$, $\gamma = 0$, subsample 0.9, column subsample 0.9, binary-logistic objective, and fixed seeds. Downstream retraining uses the same XGBoost setting for the GBDT-mechanism results; auxiliary learner slots use LightGBM, CatBoost/fallback histogram boosting, random forests, and MLPs only for robustness checks. For LightGBM we report the current setting explicitly: 120 trees, max depth 4, learning rate 0.06, 15 leaves, subsample and column subsample 0.9, and the library default minimum child samples of 20. This default constrains splits at 10–25-row budgets, so no conclusion relies on LightGBM-only tiny-budget superiority; the artifact repository includes the budget-aware sweep command for `min_child_samples` $\in \{1, 2, 5, 10, 20\}$. Hardware for the reported local runs is a CPU workstation with no GPU requirement.

### E. Reproducibility

All datasets are public, and all preprocessing is deterministic: categorical features are ordinal/one-hot encoded as specified in the scripts, numerical features are binned with shared full-data bin edges, and train/validation/test splits use stratified seeds where labels permit. The artifact repository, https://github.com/VIVAN-RANGRA/sld-gbdt-condensation (tag `icdm-repro-v1`), contains the exact configs, seeds, package versions (Python 3.11.9, NumPy 2.4.4, pandas 2.2.3, scikit-learn 1.8.0, XGBoost 3.2.0, LightGBM 4.6.0, SciPy 1.17.1), processed metric CSVs, plotting scripts, and the runbook commands for `scripts/19_run_icdm_core.py` with `configs/dataset26_binary.yaml` and seeds 0–4, `scripts/32_run_phase2_experiments.py`, and `scripts/34_run_large_scale_experiment.py`. The repository is intended to reproduce the reported tables exactly when cached processed data are used, and to closely replicate them from raw public data.

TABLE I: The 17 condensation methods. †: adapted from differentiable settings.

| Family | Method | Signal / citation |
|---|---|---|
| Geometry | Random | Uniform |
| | Herding | Moments [3] |
| | K-center | Coverage [18] |
| GBDT-native | GOSS† | Grad-norm [9] |
| | MVS† | Min-var [21] |
| Gradient | GraNd† | Norm at init [20] |
| | EL2N† | Error L2 [20] |
| | CRAIG† | Submod. [4] |
| | GradMatch† | Grad match [19] |
| | Grad. samp.† | Loss-weighted |
| Distrib. | DistributionMatching | Distr. match [10] |
| Gain / landscape | HD-Greedy | $(1-1/e)$ coverage |
| | HD-Refined | Landscape fidelity |
| | HD-Density | Land.+density |
| | Gain sketch | Gain sketch |
| | Gain-path sk. | Gain-path sketch |
| | Legacy gp | Gain-path refined |

## V. Results

### A. Does the Landscape Predict Performance?

**Finding 1: Method-level split-regret ranks method families in the small-to-mid-size benchmark.** Figure 2 and Table IV give the method-level audit. Split-regret, computed before candidate retraining, is strongly monotone with eventual AUROC at the method-mean level; gain-path methods sit in the low-regret/high-AUROC corner, while the coverage-optimal HD-Greedy method sits at the opposite extreme. Gradient sampling is the main exception, and the decomposition below shows why: its problem is leaf-value estimation, not split-structure preservation. This 17-method diagnostic grid uses 26 datasets, two audited budgets, and three seeds, and is separate from the five-budget, five-seed core grid in Table III. The large-scale audit in Section V-E bounds the claim: once high-gain margins are cleared on very large datasets, root split-regret loses rank resolution.

**Finding 2: Within a dataset, lower raw SLD predicts higher AUROC, but pooling across datasets is confounded.** Figure 3 shows the pattern directly. Within each dataset, lower raw $\delta_\infty$ corresponds to better retrained AUROC; after pooling datasets, the trend can invert because dataset scale affects both raw SLD magnitude and absolute AUROC. Raw SLD should therefore be used for within-dataset, within-budget method comparison, not as a cross-dataset absolute score.

TABLE II: Expanded 26-dataset binary benchmark inventory. OpenML entries give dataset ID/task ID when a task was used; PMLB entries use exact PMLB identifiers. $*$: focused binary suite.

| Dataset | ID | N | F | Dataset | ID | N | F |
|---|---|---|---|---|---|---|---|
| adult* | PMLB:adult | 20k | 14 | gametes_epi_1000 | PMLB:GAMETES_Epistasis_2_Way_1000atts_0.4H_EDM_1_EDM_1_1 | 1,600 | 1000 |
| agaricus_lepiota | PMLB:agaricus_lepiota | 8,145 | 22 | gametes_epi_20_0.1H | PMLB:GAMETES_Epistasis_2_Way_20atts_0.1H_EDM_1_1 | 1,600 | 20 |
| australian* | OpenML d40981 | 690 | 14 | gametes_epi_20_0.4H | PMLB:GAMETES_Epistasis_2_Way_20atts_0.4H_EDM_1_1 | 1,600 | 20 |
| bank_marketing* | OpenML d1461 | 20k | 16 | gametes_epi_3way | PMLB:GAMETES_Epistasis_3_Way_20atts_0.2H_EDM_1_1 | 1,600 | 20 |
| breast_w* | OpenML d15/t15 | 699 | 9 | gametes_het_50 | PMLB:GAMETES_Heterogeneity_20atts_1600_Het_0.4_0.2_50_EDM_2_001 | 1,600 | 20 |
| credit_approval | OpenML d29/t29 | 690 | 15 | gametes_het_75 | PMLB:GAMETES_Heterogeneity_20atts_1600_Het_0.4_0.2_75_EDM_2_001 | 1,600 | 20 |
| credit_g* | OpenML d31/t31 | 1k | 20 | hill_valley_with_noise | PMLB:Hill_Valley_with_noise | 1,212 | 100 |
| diabetes* | OpenML d37/t37 | 768 | 8 | hill_valley_without_noise | PMLB:Hill_Valley_without_noise | 1,212 | 100 |
| electricity | OpenML d151/t219 | 20k | 8 | kr_vs_kp | OpenML d3/t3 | 3,196 | 36 |
| pc1* | OpenML d1068 | 1,109 | 21 | pc3 | OpenML d1050/t3903 | 1,563 | 37 |
| pc4 | OpenML d1049/t3902 | 1,458 | 37 | phoneme | OpenML d1489 | 5,404 | 5 |
| qsar_biodeg | OpenML d1494 | 1,055 | 41 | sick | OpenML d38/t3021 | 3,772 | 29 |
| spambase* | OpenML d44/t43 | 4,601 | 57 | tic_tac_toe | OpenML d50/t49 | 958 | 9 |

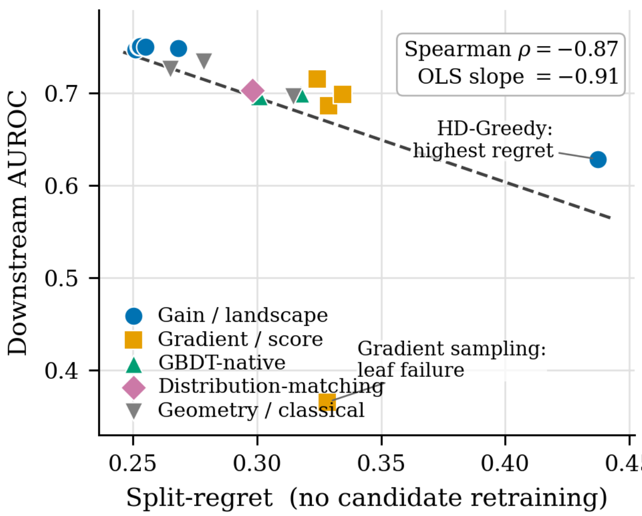


Fig. 2: **No-candidate-retraining method audit.** Each point is one condensation method. Lower split-regret, computed before candidate retraining, tracks higher downstream AUROC over 156 dataset–budget–seed cells (Spearman $\rho = -0.87$, OLS slope $-0.91$).

TABLE III: Headline performance and fidelity for five core methods (3,250 cells). AUROC uses mean±95% CI over dataset clusters; avg. rank lower is better. Speedup is relative to full-data retraining on the same dataset/learner.

| Method | AUROC | Rank | Raw $\mathrm{SLD}_\infty$ | Speedup |
|---|---|---|---|---|
| Full data | $0.812 \pm 0.062$ | – | – | $1.0\times$ |
| Legacy gain-path | $0.743 \pm 0.057$ | 2.60 | 99.4 | $3.9\times$ |
| HistDistill-Refined | $0.742 \pm 0.057$ | 2.68 | 93.6 | $3.5\times$ |
| HistDistill-Density | $0.742 \pm 0.057$ | 2.65 | 93.7 | $3.5\times$ |
| Herding | $0.731 \pm 0.054$ | 3.44 | 174.1 | $3.8\times$ |
| Random | $0.724 \pm 0.056$ | 3.63 | 138.3 | $4.0\times$ |

### B. Coverage Is Not Predictive Fidelity

**Finding 3: A near-optimal coverage selector is weak by downstream predictive performance.** HD-Greedy optimises split-threshold coverage and achieves the expected coverage win, but Figure 4 and Table V show that this objective is not the one that preserves downstream AUROC. The method lands in the bottom tier of the 17-method audit (Table IV); unweighted threshold coverage and gain-weighted landscape fidelity are different objectives.

**Finding 4: The coverage failure is structural; gradient sampling fails for a different reason.** The reference chain in Figure 5 and Table VI separates structure error from leaf-estimate gap. HD-Greedy's loss comes from choosing the wrong gain-weighted structure, not from poor leaf values after its structure is fixed. Gradient sampling is the opposite diagnostic case: its population mismatch is low, but its leaf-estimate gap is high, pointing to distorted

TABLE IV: Seventeen-method diagnostic audit. Values are method means ±95% CI over dataset clusters, over 156 dataset–budget–seed cells. Lower split-regret and higher AUROC are better.

| Method | AUROC | Regret |
|---|---|---|
| Gain-path sketch | $0.748 \pm 0.062$ | $0.251 \pm 0.079$ |
| Legacy gain-path | $0.751 \pm 0.062$ | $0.252 \pm 0.079$ |
| HD-Density | $0.751 \pm 0.061$ | $0.253 \pm 0.073$ |
| HD-Refined | $0.750 \pm 0.062$ | $0.255 \pm 0.077$ |
| Random | $0.726 \pm 0.060$ | $0.265 \pm 0.064$ |
| Gain sketch | $0.749 \pm 0.062$ | $0.268 \pm 0.081$ |
| Herding | $0.734 \pm 0.059$ | $0.279 \pm 0.082$ |
| DM | $0.703 \pm 0.062$ | $0.298 \pm 0.075$ |
| MVS | $0.697 \pm 0.058$ | $0.301 \pm 0.068$ |
| K-center | $0.697 \pm 0.062$ | $0.315 \pm 0.079$ |
| GOSS | $0.700 \pm 0.058$ | $0.317 \pm 0.073$ |
| GradMatch | $0.715 \pm 0.057$ | $0.324 \pm 0.086$ |
| Grad. samp. | $0.366 \pm 0.067$ | $0.328 \pm 0.078$ |
| EL2N | $0.686 \pm 0.056$ | $0.329 \pm 0.071$ |
| GraNd | $0.686 \pm 0.056$ | $0.329 \pm 0.071$ |
| CRAIG | $0.699 \pm 0.061$ | $0.335 \pm 0.082$ |
| HD-Greedy | $0.629 \pm 0.050$ | $0.437 \pm 0.106$ |

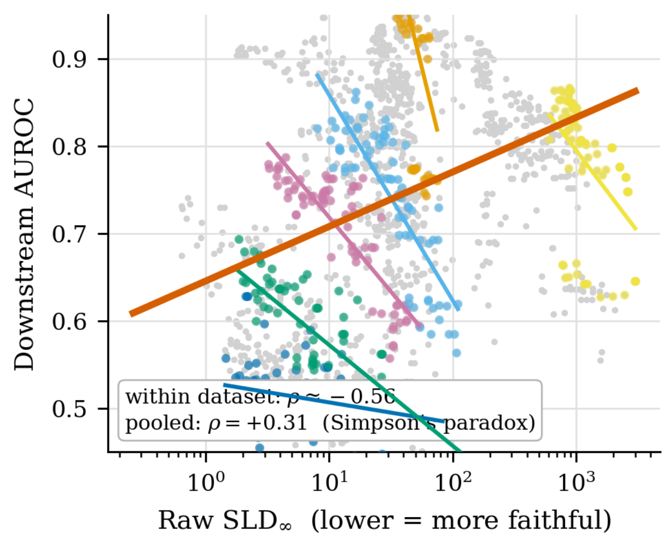


Fig. 3: **Within-dataset SLD trend.** Coloured lines are within-dataset fits on log-scale raw SLD; the average within-dataset trend is negative. The pooled fit reverses because dataset scale confounds raw SLD and AUROC.

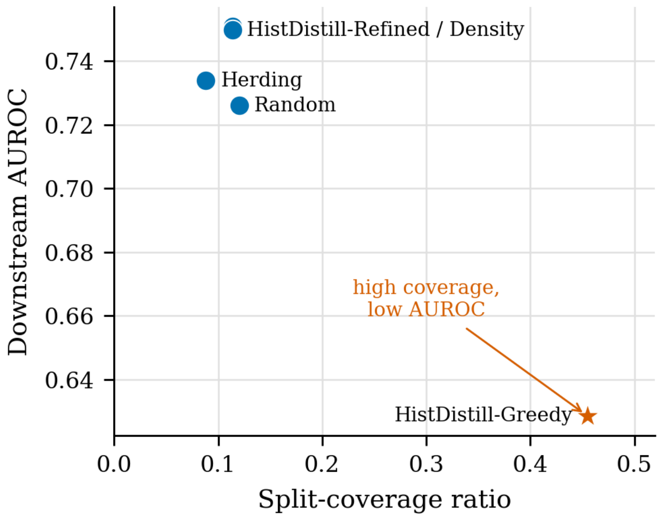


Fig. 4: **Coverage is not predictive fidelity.** HD-Greedy covers the most high-gain split thresholds, but the accurate methods occupy a different region of the coverage–AUROC plane.

TABLE V: Coverage metrics versus AUROC on the two-budget coverage audit. Cov. is high-gain split coverage; Proxy is coverage divided by the proxy upper bound.

| **Method** | **Cov25** | **Proxy25** | **Cov50** | **Proxy50** | **AUROC** |
|---|---|---|---|---|---|
| HD-Greedy | 0.371 | 1.000 | 0.539 | 1.000 | 0.629 |
| Random | 0.082 | 0.246 | 0.159 | 0.299 | 0.726 |
| HD-Refined | 0.075 | 0.223 | 0.152 | 0.286 | 0.750 |
| HD-Density | 0.075 | 0.223 | 0.152 | 0.286 | 0.751 |
| Herding | 0.059 | 0.183 | 0.118 | 0.229 | 0.734 |

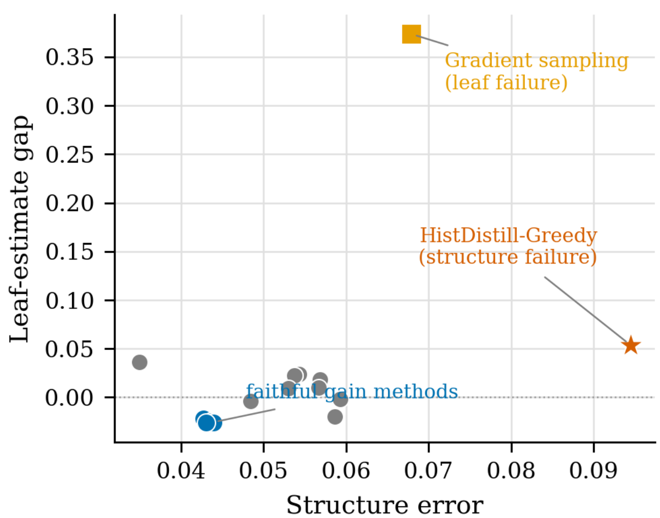


Fig. 5: **Two failure modes.** HD-Greedy is structure-dominated; gradient sampling is leaf-estimate dominated. Faithful gain methods occupy the low-error corner.

within-leaf gradient/Hessian or reweighting statistics.

TABLE VI: Error decomposition on the diagnostic subset used to illustrate both failure modes. Absolute AUROCs differ from the pooled values in Table III and Figure 2. Structure error is reference minus leaf-refit; leaf gap is leaf-refit minus deployed.

| **Method** | **Dep.** | **Leaf** | **Str. Err** | **Leaf Gap** | **Pop.** |
|---|---|---|---|---|---|
| HD-Greedy | 0.643 | 0.696 | 0.094 | 0.054 | 0.243 |
| Grad. samp. | 0.349 | 0.723 | 0.068 | 0.374 | 0.133 |
| DM | 0.720 | 0.756 | 0.035 | 0.036 | 0.235 |
| K-center | 0.713 | 0.737 | 0.054 | 0.024 | 0.196 |
| MVS | 0.724 | 0.734 | 0.057 | 0.010 | 0.088 |
| GradMatch | 0.734 | 0.732 | 0.059 | −0.002 | 0.113 |
| HD-Refined | 0.773 | 0.747 | 0.044 | −0.026 | 0.060 |
| Legacy gp | 0.774 | 0.748 | 0.043 | −0.026 | 0.068 |

Decomposition-subset anchors: full AUROC = 0.826, reference = 0.791.

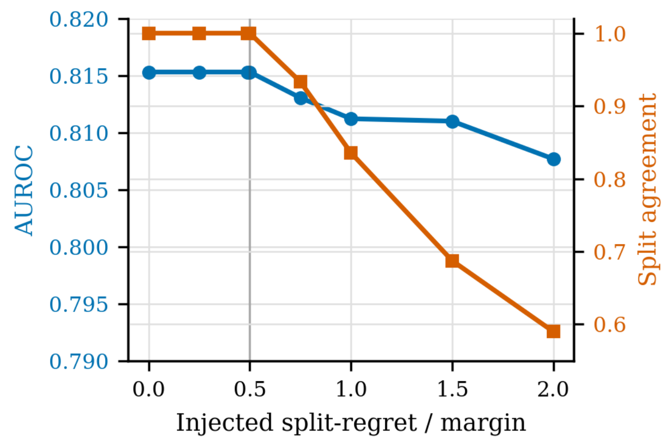


Fig. 6: **Robustness pattern.** Split agreement drops after dose 0.5; AUROC degrades mildly as calibrated split-gain noise exceeds the margin.

TABLE VII: Dose-response: 26 binary datasets, 1,872 cells. Values are mean±95% CI over dataset clusters; eight injected-noise levels are grouped into five displayed rows.

| **Dose** | **AUROC** | **ΔAUC** | **Agree.** | **Regret** |
|---|---|---|---|---|
| ≤0.50 | 0.815 ± 0.068 | 0.000 | 1.000 ± 0.000 | 0.000 |
| 0.75 | 0.813 ± 0.069 | −0.002 | 0.933 ± 0.002 | 0.011 |
| 1.00 | 0.811 ± 0.069 | −0.004 | 0.836 ± 0.004 | 0.030 |
| 1.50 | 0.811 ± 0.069 | −0.004 | 0.687 ± 0.008 | 0.069 |
| 2.00 | 0.808 ± 0.070 | −0.008 | 0.590 ± 0.011 | 0.101 |

### *C. How Much Can Structure Be Perturbed?*

**Finding 5: GBDT structure shows a margin-linked robustness pattern.** Figure 6 and Table VII inject calibrated split-gain noise into the full-data landscape. AUROC and split agreement stay flat while the margin condition in (3) holds; beyond that point, agreement drops first and AUROC degrades mildly. The result explains why aggressive condensation can work when the dominant split margins are preserved, and why corrupting high-gain structure is disproportionately costly.

### *D. Performance, Fidelity, and Cost*

**Finding 6: HistDistill-Refined and -Density are practically tied with the strongest prior selector while being the most faithful.** Tables III and X carry

TABLE VIII: Large-scale audit at the largest training size per dataset. $\rho$ is Spearman correlation over the five audited methods; negative $\rho$ means lower split-regret aligns with higher AUROC. Each row is a single large-scale run.

| Dataset | Rows | $\rho$ | Top speedup |
|---|---|---|---|
| Covertype | 0.48M | −0.36 | 160× |
| SUSY | 4.50M | −0.97 | 1,864× |
| HIGGS | 10.50M | 0.11 | 5,734× |

the performance, fidelity, and cluster-aware paired comparison numbers. The interpretation is practical parity at the top: the refined family matches the legacy gain-path selector in AUROC while achieving the lowest landscape discrepancy among the core methods. The Refined-vs.-Legacy mean difference is operationally negligible and its dataset-clustered confidence interval crosses zero. At the studied small-data budgets, condensed retraining gives a median 3–4× speedup, with median condensed training under half a second in the recorded runs; SLD itself costs one histogram pass.

### *E. Does the Diagnostic Hold at Scale?*

**Scale audit: the cost payoff grows with $N$, while ordering is dataset-dependent.** Figure 7 and Tables VIII and IX report a real large-data audit using 400-row condensed sets. Retrain speedup rises from single digits at 20k rows to hundreds or thousands at million-row scale. Random has the best large-$N$ AUROC point estimate on all three datasets, but the three-dataset paired audit is not powered to claim statistical superiority: Random's mean advantage over HD-Density is 0.034 AUROC and over HD-Refined is 0.035 AUROC, with two-sided Wilcoxon $p = 0.25$ in both comparisons. The practical reading is convergence among reasonable condensers, not a new winner-take-all method ranking. This is the regime thesis in one comparison: HD-Refined beats Random in 24/26 small-to-mid-size datasets in the clustered audit (Table X), yet Random has the largest point estimate on all three very large datasets (Table IX).

The HIGGS case is informative rather than contradictory. Several methods have zero root split-regret there, yet their AUROCs differ. This is exactly the margin story from Figures 3 and 6: at very large $N$, the dominant root split is so well separated that many surrogates clear the same margin, and leaf-estimate quality or deeper structure becomes the discriminating factor. Root-level regret remains useful for rejecting severe failures, but not for ranking reasonable surrogates once the top split is saturated. A rule that only rejects the obvious gradient-sampling failure retains the winner in all scale cells while saving less than 2% of already-small condensed-search time; the real cost win comes from training on hundreds of rows rather than millions.

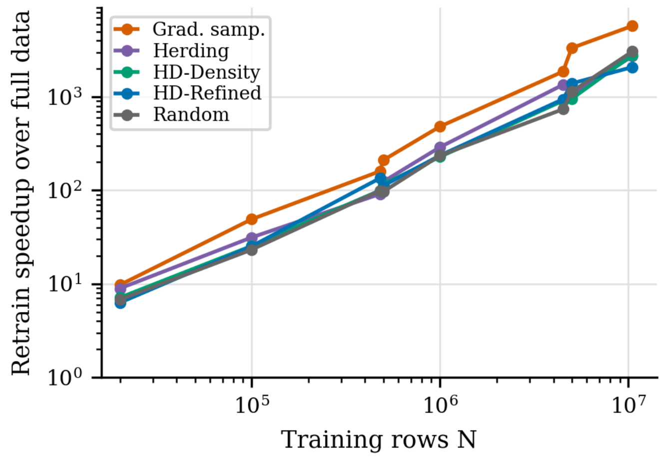


Fig. 7: **Large-scale retraining payoff.** Log-log speedup for 400-row condensed retraining on Covertype, SUSY, and HIGGS. Points are single runs at observed dataset sizes, so no run-to-run CI is claimed for this scale audit.

TABLE IX: Large-scale AUROC at the largest training size. Full-data AUROCs are 0.897, 0.873, and 0.803 for Covertype, SUSY, and HIGGS. Random wins the condensed point estimates, but the three-dataset paired audit is underpowered ($p = 0.25$ against HD-Refined and HD-Density); no per-row CI is claimed because the large-scale audit uses one full run per dataset/method.

| Method | Cov. | SUSY | HIGGS |
|---|---|---|---|
| Full data | 0.897 | 0.873 | 0.803 |
| Random | **0.792** | **0.839** | **0.687** |
| HD-Refined | 0.747 | 0.818 | 0.648 |
| HD-Density | 0.781 | 0.794 | 0.641 |
| Herding | 0.719 | 0.780 | 0.450 |
| Grad. samp. | 0.377 | 0.219 | 0.364 |

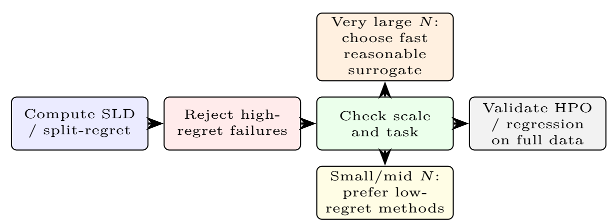


Fig. 8: **Operational workflow implied by the measurements.** SLD is most useful as a cheap screening diagnostic: reject high-regret failures, prefer low-regret methods only when margins are not saturated, and validate HPO, regression, and final deployment choices on full data.

TABLE X: Dataset-clustered core-method comparisons. $\Delta$ is the paired mean AUROC difference over 3,250 cells; confidence intervals bootstrap the 26 dataset clusters.

| Method | Comparator | $\Delta$ | 95% CI | DS wins |
|---|---|---|---|---|
| HD-Refined | Legacy gp | −0.00056 | [−0.0019, 0.0008] | 11/26 |
| HD-Density | Legacy gp | −0.00056 | [−0.0016, 0.0005] | 9/26 |
| HD-Refined | Herding | 0.01183 | [0.0037, 0.0204] | 18/26 |
| HD-Refined | Random | 0.01807 | [0.0124, 0.0233] | 24/26 |

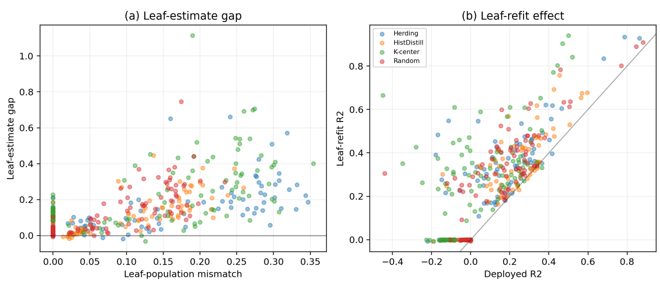


Fig. 9: **Regression failure modes.** Left: leaf-population mismatch tracks leaf-estimate loss. Right: refitting leaves on full data helps, but the remaining structure gap dominates.

## VI. Where the Lens Stops: Scope and Limitations

The SLD diagnostic is designed around the inductive bias of histogram GBDTs. Three scopes delimit where that design leads to weak or negative results.

**Regression is structure-dominated, and no condenser fixes it.** The regression audit uses eight public datasets (diabetes, California housing, abalone, cpu_act, kin8nm, house_16H, elevators, wine quality), budgets 25/50, seeds 0–2, and XGBoost/LightGBM/MLP downstream learners. SLD is recomputed for squared-error gradients. The structure reference–the full-data tree structure with leaves refit under the decomposition protocol–achieves $R^2 \approx 0.63$. The leaf-refit model, which uses each condensed set's structure but refits leaves on full data, falls to $R^2 = 0.28$–$0.29$. The deployed condensed models then reach only $R^2 \approx 0.16$ for HistDistill and 0.05 for K-center. Thus, for HistDistill, the total gap from the structure reference to the deployed model is about 0.47; of that gap, about 0.35 is structural $(0.63 - 0.28)$ and about 0.12 is due to leaf-value estimation $(0.28 - 0.16)$. No method we evaluate closes the structure gap: the best deployed condensed $R^2$ is 0.16, well below even the leaf-refit baseline of 0.28. Figure 9 shows this pattern. The SLD diagnostic correctly *diagnoses* this failure, since the condensed sets have high $\mathrm{SLD}_\infty$ relative to full-data regression baselines, but it does not cure it. Regression tasks appear to require structure fidelity that current condensers cannot achieve at the budgets we study.

**HPO transfer is weak.** We test whether condensed sets preserve hyperparameter rankings on 12 binary datasets using HistDistill-Refined at budget 50. Each dataset evaluates the same 100 XGBoost hist configurations on the condensed and full train/validation splits; the random-search space is 50–150 trees, depth 2–5, learning rate 0.025–0.18, min-child-weight 0.5–8, subsample/column-subsample 0.65–1, $\lambda = 0.05$–$8$, and $\gamma = 0$–$2$. Across datasets, top-3 Jaccard between condensed- and full-data HPO results is 0.042, top-1 match rate is 0.083, and Spearman rank correlation is 0.145. Median HPO speedup is 1.57×. The mean final test-AUROC gap is only 0.00125, but that is because condensed HPO finds a different competitive configuration, not because it transfers the full-data ranking. Condensed HPO should therefore be followed by full-data checks of top configurations before deployment.

**No-retraining selection of the single best method remains weak.** SLD ranks method families well at the aggregate level (Spearman −0.87), but selecting the *single best* method for a new dataset based only on SLD is harder. The top-3 success rate of the SLD-based selector is 0.436 across 156 dataset–budget–seed cells, and the top-1 rate is 0.122. The split-regret selector performs similarly. The landscape score is useful for screening out bad condensers and ranking families, but the gap between the top-2 or top-3 methods is often smaller than the within-dataset variance of SLD, making single-method selection unreliable. Future work could combine the landscape score with dataset-level meta-features to improve per-instance selection.

Multiclass classification is a thin positive extension (HistDistill 0.910 macro-AUROC vs. Random 0.897), but taken together these scopes mark the boundary of the diagnostic: SLD is built around the GBDT split-gain objective, and its predictive value erodes as a task drifts away from that inductive bias.

## VII. Conclusion

Measuring mechanism, not just downstream predictive performance, reveals things that a retrain-and-compare protocol cannot. For gradient-boosted decision trees on tabular data, the split-gain landscape is a central mechanism for tree-structure selection: it determines which features and thresholds the booster commits to before a leaf value is estimated. Comparing landscapes between a full dataset and its surrogate, through the Split-Landscape Discrepancy, yields a no-candidate-retraining diagnostic whose method-level split-regret correlates with downstream AUROC in the small-to-mid-size benchmark at Spearman −0.87. The scale audit qualifies this claim: once very large datasets make the dominant split margins easy to preserve, random sampling becomes difficult to beat and root split-regret becomes a screening signal rather than a fine-grained ranker.

The most practically consequential finding is that *coverage is not predictive fidelity.* A condensation method with a provable near-optimal guarantee on unweighted split-threshold coverage is simultaneously among the weakest downstream performers in the pooled audit. The reason is structural: unweighted coverage fills the landscape broadly, while AUROC depends on gain-weighted, Hessian-weighted, path-sensitive structure preservation. The reference-chain decomposition makes this precise: the coverage-optimal method's failure is a structure failure rather than a leaf-value failure.

We also delimit the scope of these findings. Regression tasks are structure-dominated in a way that current condensers cannot overcome. Hyperparameter transfer from condensed sets to full-data rankings is weak. At deployment scale, reasonable condensers can converge in root split-regret while still differing in leaf estimates and deeper structure. These boundaries reinforce the role of SLD as a measurement instrument rather than just a performance metric.

The applied workflow is therefore direct: before choosing a condensation strategy, compute SLD in a histogram pass and use split-regret to reject bad candidates. This screens out poor surrogates before retraining on every candidate; when $N$ is very large and several candidates clear the dominant split margins, choose among the reasonable surrogates by budget, speed, and downstream validation. Coverage metrics should not be used as an AUROC proxy, and for regression tasks or HPO transfer condensed sets should be treated as approximations to validate rather than substitutes for the full data. We release the diagnostic and benchmark in the single-blind artifact repository to make this workflow reproducible and extensible to new condensation methods as they emerge.

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